

Questions for the Audience

Which Activation Functions are Commonly used as $g^{(1)}(a)$?

- Sigmoid activation: $\sigma(a)$ or $\tanh(a)$
- Rectified Linear Units (ReLU) and its variants, see Lecture 6
- Gaussian activation, see RBF networks in Lecture 12

What can we do when the Learning Diverges?

- Decrease the Learning Rate
- Include momentum learning
- Normalize/Standardize the inputs
- Initialize the weights with Xavier method
- Decrease the number of hidden units K

Questions for the Audience

How many Outputs can a Binary Classification Network have

- $O = 1$ with σ activation and $\mathcal{J}^{\text{BCE}}$
- $O = 2$ with Softmax activation and $\mathcal{J}^{\text{CCE}}$
- $O > 2$ also possible with $\mathcal{J}^{\text{CCE}}$, other outputs are ignored

What Happens to SoftMax Outputs under the Following Conditions

- All logit values are identical: $\forall o : z_o = c$
→ All probabilities are equal $\forall o : y_o = 1/O$
- We add a constant c to all values: $\forall o : z'_o = z_o + c$
→ Nothing! $\forall o : y'_o = y_o$
- We multiply a constant $0 \leq c < 1$ to all values: $\forall o : z'_o = z_o \cdot c$
→ Probabilities y'_o are more similar to each other than the original y_o